

Subject Description Form

Subject Code	AAE4202
Subject Title	Electronics & Information Technologies for Unmanned Aerial Systems
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AAE2005 Electrics and Electronic for Aeronautical Engineering
Objectives	To provide students with knowledge of electronics and information technologies for unmanned aerial vehicles (UAV) and unmanned aircraft systems (UAS).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Possess all required concepts and skills related to the remote control and primary autonomous unmanned aircraft systems; and - Apply the learnt concepts and skills to operate, maintain and perform diagnosis on existing unmanned aircraft systems; and - Extend their knowledge to analyse and develop new modules or algorithms in unmanned aircraft systems for desired needs.
Subject Synopsis/ Indicative Syllabus	System Component, Electronic Device, and Radio Link Common system components of UAS: airframe, servo, propulsion system (motor, electronic speed controller (ESC), propeller), Li-po battery, radio transmitter and receiver, telemetry, ground control station (GCS), and the autopilot. Dynamic Modelling of Unmanned Aerial Vehicle Coordinate systems, kinematic model, dynamic model, propulsion system model, controller allocation model of UAS and model linearisation method. Flight Control Framework Cascade control structure, position control, attitude control, and control allocation for the low-level control of UAS. Path and Trajectory Planning Global path planning for UAS including search-based methods and sample-based methods. Local smooth trajectory generation methods. Autopilot System Integration and Flight Simulation Open-source flight controller; Flight simulation platform; Programming and hardware interface; Implementation of control and planning algorithms, Introduction to autonomous aerial robotic system. Mini UAS Flight Test Integrate the hardware and software; flight test in an indoor laboratory.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures/hands on sessions, assignments, test, project and examination. 2. The continuous assessment and examination are aimed at providing students with integrated knowledge required for unmanned aircraft systems. 3. Technical/practical examples and problems are raised and discussed in class/hands on sessions. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Intended subject learning outcomes to be covered</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Lecture</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Hands on</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Assignment</td><td>✓</td><td>✓</td><td></td></tr><tr><td>4. Project</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes to be covered			a	b	c	1. Lecture	✓	✓		2. Hands on	✓	✓		3. Assignment	✓	✓		4. Project	✓	✓	✓										
Teaching/Learning Methodology	Intended subject learning outcomes to be covered																																	
	a	b	c																															
1. Lecture	✓	✓																																
2. Hands on	✓	✓																																
3. Assignment	✓	✓																																
4. Project	✓	✓	✓																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Assignment</td><td>20 %</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. In-class test</td><td>20 %</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Project (flight show and project report)</td><td>50 %</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Presentation</td><td>10 %</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 1.0 × Continuous Assessment The continuous assessment consists of four components: assignments, in-class test, project and presentation. The assignments and the in-class tests aim at enhancing students’ understanding of the lecture content. The project enables students to apply the learnt knowledge in lectures to practice. Students will learn the basic hardware components for UAS, implement some algorithms learnt in lectures, integrate the algorithms in the provided hardware and conduct flight test	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Assignment	20 %	✓	✓		2. In-class test	20 %	✓	✓		3. Project (flight show and project report)	50 %	✓	✓	✓	4. Presentation	10 %	✓	✓	✓	Total	100%			
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed																														
		a	b	c																														
1. Assignment	20 %	✓	✓																															
2. In-class test	20 %	✓	✓																															
3. Project (flight show and project report)	50 %	✓	✓	✓																														
4. Presentation	10 %	✓	✓	✓																														
Total	100%																																	

	in an indoor laboratory. Completing the project indicates that the students grasp the learnt knowledge. In the presentation, students will present selected topics relevant to the subject content to enhance their understanding.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	21 Hrs.
	▪ Project	18 Hrs.
	Other student study effort:	
	▪ Self-study and extended reading	22 Hrs.
	▪ Practice	44 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. Quan, Quan. Introduction to multicopter design and control. Springer, 2017 2. Kenzo Nonami et al, Autonomous flying robots: unmanned aerial vehicles and micro aerial vehicles, Springer, 2010. 3. Donald Norris, Build your own quadcopter: power up your designs with the Parallax Elev-8, New York: McGraw-Hill Education, 2014	

Oct 2025